

#	Category	Item / Function
MOTOR & DRIVE		
1		AC Motor (bench test unit)
		VFD (Variable Frequency Drive)
2		
		Dynamic Braking Resistor
3		
POWER DISTRIBUTION		
4		Main Power Switch
		VFD Branch Breaker
5		
		DC Power Supply
6		
		Main Power Contactor
7		
8		Fuse Block - 24V Button Common
9		Fuse Block - AC Branches
PLC CONTROL SYSTEM		
10		PLC CPU / Ethernet Module
11		PLC Rack / Base
12		Discrete Input Module (24 VDC)
13		Analog Current Input Module
		RTD Input Module
14		
		Relay Output Module
15		
		Analog Voltage Output Module
16		
17		PLC Rack Filler Panels
HMI / OPERATOR DISPLAY		
18		C-More Micro Touch Panel
OPERATOR CONTROLS		
19		Forward Pushbutton Head
20		Reverse Pushbutton Head
21		Deadman/Run Pushbutton Head
22		Reset Pushbutton Head
		NO Contact Block (1NO finger-safe)
23		
24		LED Block - Green (24V DC)
25		LED Block - Yellow (24V DC)
26		LED Block - Blue (24V DC)
27		LED Block - Red (24V DC)
		Emergency Stop Pushbutton
28		
29		Panel Indicator Light - Error/Fault
MOTOR CONNECTIONS		
30		Banana Socket - Blue (4 mm)

- 31 Banana Socket - White (4 mm)
- 32 Banana Socket - Red (4 mm)

WIRE, DUCT, & HARDWARE

- 33 Slotted Wire Duct (Type G, 1" x 1")
- 34 Duct Cover for G1X1LG6
- 35 Insulated Crimp Ferrules, 12 AWG (gray)
- 36 Control Wire - 18 AWG stranded Blue
- 37 Return Wire - 18 AWG White or Blue/W
- 38 Power Wire - 12 AWG Black
- 39 Ground Wire - 12 AWG Green/Yellow
- Fuse - 500 mA fast-blow (5x20 mm)
- 40
- 41 Fuse - 1 A fast-blow (5x20 mm)

Manufacturer	Model / Part Number	Qty	Unit
GE	5KE182BC205B	1	EA
Huanyang	HY02D211B-T	1	EA
ATO	APCS-300R30-AD	1	EA
Powertec	71008	1	EA
Phoenix Contact	TMC 8 3C 15A (P/N 2907618)	1	EA
Mean Well	NDR-480-24	1	EA
Mitsubishi	SD-N35	1	EA
Morsettitalia	EURO S4LH	1	EA
Morsettitalia	EURO S10H-5H	1	EA
AutomationDirect	H2-DM1E	1	EA
AutomationDirect	D2-09B-1	1	EA
AutomationDirect	D2-08ND3	1	EA
AutomationDirect	F2-08AD-1	1	EA
AutomationDirect	F2-04RTD	1	EA
AutomationDirect	D2-08TR	1	EA
AutomationDirect	F2-08DA-2	1	EA
AutomationDirect	D2-Fill	3	EA
AutomationDirect	EA1-T4CL	1	EA
Autonics	S2PR-P3G	1	EA
Autonics	S2PR-P3Y	1	EA
Autonics	S2PR-P3B	2	EA
Autonics	S2PR-P3R	1	EA
Autonics	SA-CA	5	EA
Autonics	SA-LDG	1	EA
Autonics	SA-LDY	1	EA
Autonics	SA-LDB	2	EA
Autonics	SA-LDR	1	EA
Cutler-Hammer / Eaton	E22B1	1	EA
Wamco	525	1	EA
Generic	Banana Socket Blue	1	EA

Generic	Banana Socket White	1 EA
Generic	Banana Socket Red	1 EA
Panduit	G1X1LG6	6 FT
Panduit	C1LG6	6 FT
Z+F	V30AE000055	500 EA
Various	18 AWG Blue stranded	1 Roll
Various	18 AWG White stranded	1 Roll
Various	12 AWG Black	1 Roll
Various	12 AWG Green/Yellow	1 Roll
Bussmann or Littelfuse	GMA-500mA or 0217.500MXP	1 EA
Bussmann or Littelfuse	GMA-1A or equiv.	1 EA

Notes / Source

3 HP, 230V 1750 RPM 4-pole TEFC. See motor/Motor_Nameplate.md.

2.2 kW (3 HP), 110V single-phase in, 220V 3-phase out. See vfd/Programmed_Parameters.md.

300 W, 30 ohm, aluminum-clad. P+/PR terminals on VFD. See vfd/Brake_Resistor_Wiring.md.

120 V ON/OFF disconnect. Powers entire control panel.

15 A thermal-magnetic DIN-rail breaker on VFD neutral leg (per as-built; reviewing for final).

480 W, 24 VDC, DIN-rail. Powers PLC, HMI, indicator LEDs, VFD control loops.

3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC.

Holds 1 A fast-blow fuse for shared pushbutton +24V supply.

Holds AC line branch fuses for PSU and PLC mains feeds.

Do-more CPU with built-in Ethernet, fits DL205 rack.

9-slot DL205 base, AC powered.

8-ch DI, sourcing mode. Slot 2. Inputs X0-X7.

8-ch analog input. Slot 3. CH1+ reads VFD VO motor current monitor.

4-ch RTD input. Slot 4. Installed but unused (spare for future thermal sensing).

8-ch relay output. Slot 5. Sinking mode. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5).

8-ch voltage output. Slot 6. +V1 drives VFD VI speed reference (0-10V).

Slots 7-9 fillers.

4-inch color touch panel HMI. Ethernet to PLC.

22 mm illuminated green pushbutton (Forward command -> X0).

22 mm illuminated yellow pushbutton (Reverse command -> X1).

22 mm illuminated blue pushbuttons (Run 1 -> X2, Run 2 -> X3).

22 mm illuminated red pushbutton (planned Reset -> X6).

One per pushbutton (was MC22-UA legacy P/N). 110VAC/10A, 220VAC/5A.

Forward button illumination -> Y0.

Reverse button illumination -> Y1.

Run 1 and Run 2 button illumination -> Y2, Y3.

Reset button illumination (planned).

22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.

28 VDC, 0.6 W panel-mount LED indicator. Driven by Y4.

Quick-disconnect for VFD U output to motor T1.

VFD V output to motor T2.

VFD W output to motor T3.

Light gray, 6 ft lengths. Panel wire management.

Light gray, snap-on cover.

10 mm length, DIN color code. Per pack.

DC control circuits per NFPA 79 §13.2.

DC return circuits per NFPA 79 §13.2.

Motor leads (U/V/W) and AC mains.

Equipment grounding per NFPA 79.

Originally spec'd for VFD jam signal +24V; per as-built note, not currently installed (review).

Pushbutton common +24V supply protection (in EURO S4LH).

Item / Function	Manufacturer	Model / Part Number
Motor PE145TC-2-4	Various	PE145TC-2-4
Circuit Breaker NBE7	Various	NBE7
30 A Branch Fuse (TBD)	Various	30 A fuse
DC Power Supply 1866-1441-ND	Digi-Key vendor	1866-1441-ND
VEVOR 3HP 120V VFD	VEVOR	VEVOR 3HP 120V VFD
ATO 3HP 120V VFD	ATO	ATO 3HP 120V VFD
110V 20A 3HP VFD (generic)	Generic	110V 20A 3HP VFD
WEG Single Phase Motor	WEG	WEG Single Phase 3HP
Worldwide Electric NT3-18-213T	Worldwide Electric	NT3-18-213T
VEVOR High Torque Motor	VEVOR	VEVOR 5HP 230V
TECO L510 VFD	TECO	L510 (2HP and 3HP)
Marathon G0532 Gearmotor	Marathon	G0532
KB Electronics KBWC-15K	KB Electronics	KBWC-15K
Capacitive Touch Sensor TTP223B	Various	TTP223B
Capacitive Touch Sensor CAP1293	Microchip	CAP1293-1-SN
Capacitive Proximity Sensor 1864-1032-ND	various	1864-1032-ND
Magnetic Non-Contact Safety Sensor F3S-TGR	Omron	F3S-TGR-NMPC-21-10
RFID Safety Sensor RSS 36-I2-D	Schmersal	RSS 36-I2-D-R-ST5
Safety Light Curtain F3SG-SR / PG	Omron	F3SG-SR / PG
Ichor-provided Red/Green Lights	Ichor (sponsor)	Ichor-provided
Ichor-provided E-stop	Ichor (sponsor)	Ichor-provided

From BOM File	Reason Not Used
BOM_02_20_26	Not used - GE 5KE182BC205B used instead for bench testing
BOM_1_29 / BOM_02_20_26	Never purchased; Phoenix Contact TMC 8 3C 15A used instead
BOM_1_29	Never needed in final design
BOM_1_29	Superceded by Mean Well NDR-480-24
VFD_120V_3HP_Decision_Matrix	Rejected during selection (hobby-grade, limited documentation)
VFD_120V_3HP_Decision_Matrix	Too expensive (\$450-500) for project budget
VFD_120V_3HP_Decision_Matrix	Considered but Huanyang HY02D211B-T selected
Motor Spec	Considered but GE 5KE182BC205B used for bench
Motor Spec	Considered but not purchased
Motor Spec	Considered but not purchased
Motor Spec	Considered but Huanyang selected
Motor Spec	Considered but not purchased
Motor Spec	Considered (triac speed control) but VFD approach used instead
BOM_1_30_beta (safety draft)	Does not meet industrial standards
BOM_1_30_beta (safety draft)	Chip only, needs interface board
BOM_1_30_beta (safety draft)	Too expensive (\$109.25)
BOM_1_30_beta (safety draft)	Too expensive; out of stock at retailers
BOM_1_30_beta (safety draft)	Suspected non-retail / industrial-only
BOM_1_30_beta (safety draft)	Out of scope for capstone budget
BOM_02_20_26	Not applicable - Wamco 525 used as fault light; button LEDs provide ready/run indication
BOM_02_20_26	Replaced with Cutler-Hammer/Eaton E22B1 procured by team